

QISKIT: MOCK MEASUREMENT OF A UNITARY 1-QUBIT OBSERVABLE

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How do we describe the measurement of a 1-qubit observable $\mathcal{O}$ (other than Pauli-Z) within the confines of the quantum circuit model?

Assuming $\mathcal{O}$ is unitary, this can always be done. A typical example of such an observable is the spin operator along the v -axis i.e. $\mathbf{v} \cdot \boldsymbol{\sigma} = \sum_{i=1}^3 v_i \sigma_i$, where v_i are the components of a unit vector in $\mathbb{R}^3$, and σ_i are the Pauli matrices. We will also assume that $\mathcal{O}$ is nondegenerate (and can thus distinguish between two possible measurement outcomes), i.e. $\mathcal{O} \neq \pm I$. Note that

$$(0.1) \quad P^\dagger \mathcal{O} P = Z$$

where $Z = \sigma_3$ (Pauli-Z matrix), and P is a matrix whose first and second columns are respectively given by normalised eigenvectors $\boldsymbol{\lambda}_1, \boldsymbol{\lambda}_{-1}$ corresponding to eigenvalues $+1$ and -1 of $\mathcal{O}$ (see Remark 0.1). Suppose we wish to take a measurement of $\mathcal{O}$ for a qubit in state $|\psi\rangle = a|0\rangle + b|1\rangle$. Let's rewrite $|\psi\rangle$ in terms of the eigenbasis of $\mathcal{O}$.

$$\begin{aligned} |0\rangle &= r|\boldsymbol{\lambda}_1\rangle + s|\boldsymbol{\lambda}_{-1}\rangle \\ |1\rangle &= u|\boldsymbol{\lambda}_1\rangle + t|\boldsymbol{\lambda}_{-1}\rangle \end{aligned}$$

which in the matrix representation, is understood as

$$I = P \begin{pmatrix} r & u \\ s & t \end{pmatrix} = P P^\dagger$$

where we used that P is unitary (since its columns are an orthonormal basis with respect to the Hermitian inner product). Thus, $P^\dagger |\psi\rangle = (ar + bu)|\boldsymbol{\lambda}_1\rangle + (as + bt)|\boldsymbol{\lambda}_{-1}\rangle = a'|\boldsymbol{\lambda}_1\rangle + b'|\boldsymbol{\lambda}_{-1}\rangle$. In component form,

$$(0.2) \quad P^\dagger \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a' \\ b' \end{pmatrix}$$

Note that (0.2) can be viewed in two different ways.

1. As a change of basis. That is, $(a, b)^T$ gives the components of $|\psi\rangle$ with respect to the eigenbasis of Z (i.e. the computational basis). Similarly, $(a', b')^T$ gives the components of $|\psi\rangle$ with respect to the eigenbasis of $\mathcal{O}$.
2. As a physical evolution of the state $|\psi\rangle$, given by unitary operator $P^\dagger$. So, $(a', b')^T$ gives the components of $P^\dagger |\psi\rangle$ with respect to the computational basis.

This equivalence is key. It tells us that: *the amplitudes of $P^\dagger |\psi\rangle$ in the computational basis are precisely the amplitudes of $|\psi\rangle$ in the eigenbasis of $\mathcal{O}$* . Let $j \in \{\pm 1\}$. In practical terms, the probability of measuring outcome j for observable $\mathcal{O}$ given state $|\psi\rangle$ is the same as for measuring outcome j for observable Z given state $P^\dagger |\psi\rangle$. Put a bit more elegantly,

$$(0.3) \quad \langle Z \rangle_{P^\dagger \psi} = \langle \psi | P Z P^\dagger | \psi \rangle \stackrel{(0.1)}{=} \langle \psi | \mathcal{O} | \psi \rangle = \langle \mathcal{O} \rangle_\psi$$

In circuit notation,

$$|\psi\rangle \text{ --- } \boxed{P^\dagger} \text{ --- } \boxed{\text{0/1}} = |\psi\rangle \text{ --- } \boxed{\mathcal{O}}$$

Remark 0.1 (Freedom in choice of P). Note that a matrix of normalised eigenvectors (with columns in a fixed order) is not unique: since each column has a ± 1 scaling freedom, there are 2^n possibilities for rank n . In the case of P then, there are 4 choices.

Example 0.2 (Mock measurement of spin along the v -axis in Qiskit). In the following, we simulate measurement of a qubit with respect to observable $\mathcal{O} = \mathbf{v} \cdot \boldsymbol{\sigma}$, noting that the fixed observable (as per the quantum circuit model) is $\mathbf{v} = (0, 0, 1)^T$, i.e. Pauli-Z. As explained above, this can be achieved by applying $P^\dagger$ immediately before measurement in the computational basis. In the following examples, we can always find a solution for P in the form a gate $R_y(\theta)$.¹

(i) $\mathbf{v} = (1, 0, 0)^T$. $\mathcal{O} = X$ is diagonalised by the Hadamard matrix. That is, $P = H$. Applying Remark 0.1 and rescaling the second column by -1 , another possibility for P is $R_y(\pi/2)$.

(ii) $\mathbf{v} = \frac{1}{\sqrt{2}}(-1, 0, -1)^T$. This gives $\mathcal{O} = -H$, whence for $\nu_\pm = \sqrt{2(2 \pm \sqrt{2})}/2$, have

$$P = \begin{pmatrix} -\nu_- & \nu_+ \\ \nu_+ & \nu_- \end{pmatrix} = \begin{pmatrix} -\sin \gamma & \cos \gamma \\ \cos \gamma & \sin \gamma \end{pmatrix}, \quad \gamma = \pi/8$$

That is, $P = R_y(\pi/4)X$, which gives $P^\dagger = XR_y(-\pi/4)$. The gate P can be constructed using either of these compositions since it is Hermitian.² Furthermore, there are 3 other possibilities for P (Remark 0.1). For instance, we could rescale the first column by a factor of -1 (call this matrix P'), where

$$P' = \begin{pmatrix} \nu_- & \nu_+ \\ -\nu_+ & \nu_- \end{pmatrix} = \begin{pmatrix} \cos \gamma' & -\sin \gamma' \\ \sin \gamma' & \cos \gamma' \end{pmatrix}, \quad \gamma' = -3\pi/8$$

That is, $P' = R_y(-3\pi/4)$. Take care to note that P' is *not* Hermitian! P and P' are equivalent (up to a global sign) to the other two solutions, so we're done.

(iii) $\mathbf{v} = \frac{1}{\sqrt{2}}(-1, 0, 1)^T$. This gives

$$\mathcal{O} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}, \quad P = \begin{pmatrix} -\nu_+ & \nu_- \\ \nu_- & \nu_+ \end{pmatrix} = \begin{pmatrix} -\sin \gamma & \cos \gamma \\ \cos \gamma & \sin \gamma \end{pmatrix}, \quad \gamma = 3\pi/8$$

That is, $P = R_y(3\pi/4)X = XR_y(-3\pi/4)$. Rescaling as in (ii), another possibility is

$$P' = \begin{pmatrix} \nu_+ & \nu_- \\ -\nu_- & \nu_+ \end{pmatrix} = \begin{pmatrix} \cos \gamma' & -\sin \gamma' \\ \sin \gamma' & \cos \gamma' \end{pmatrix}, \quad \gamma' = -\pi/8$$

That is, $P' = R_y(-\pi/4)$. Additional comments for these calculations follow the same format as in (ii).

Remark 0.3 (Rotation angle in Bell experiment). We'll follow the presentation of Bell's inequality from Section 2.6 of Nielsen & Chuang [1].

$$|\beta_{11}\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}, \quad Q = Z, \quad R = X, \quad S = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix} \cdot \boldsymbol{\sigma}, \quad T = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \cdot \boldsymbol{\sigma}$$

Measurements are taken of $Q \otimes S$, $R \otimes S$, $R \otimes T$ and $Q \otimes T$ with respect to a state $|\psi\rangle$ (reproduced for each measurement). We get that

$$(0.4) \quad \langle Q \otimes S \rangle_\psi + \langle R \otimes S \rangle_\psi + \langle R \otimes T \rangle_\psi - \langle Q \otimes T \rangle_\psi = 2\sqrt{2}$$

for $|\psi\rangle = |\beta_{11}\rangle$. This violates Bell's inequality (in fact, maximally, by a result of Tsirelson)³. Note that $R(\theta/2) = R_y(\theta)$, where R is the canonical rotation matrix in $\mathbb{R}^2$. The half-angle is explained by the following degeneracy: $R(\alpha + \pi) = -R(\alpha)$, so it suffices for the argument of R to be $\alpha \in [0, \pi]$. Using the arguments above (and Example 0.2), we can mock-up observables $\mathcal{O} = Q, R, S, T$ by applying $R_y(2\alpha_{\mathcal{O}})$ as follows. The angles $\alpha_{\mathcal{O}} \in [0, \pi]$ are given by

$$(0.5) \quad \alpha_Q = 0, \quad \alpha_R = -\pi/4, \quad \alpha_S = 3\pi/8, \quad \alpha_T = \pi/8$$

¹In an optical system, such a gate is closely related to a beam-splitter.

²Using P Hermitian, we see the circuit identity $XR_y(\theta)X = R_y(-\theta)$ in action.

³The expectation value (while varying over any observables of the form $\mathbf{v} \cdot \boldsymbol{\sigma}$ and states $|\psi\rangle$) is bounded by $2\sqrt{2}$, and $|\beta_{11}\rangle$ is an example of a state that saturates this bound for the above choices of Q, R, S, T .

Exercise 0.4. Denote the expression on the left-hand side of (0.4) by $\mathbf{E}(\psi)$.

- (i) Determine conditions on Q, R, S, T such that $\mathbf{E}(\psi)$ does not violate Bell's inequality for any choice of $|\psi\rangle$, i.e. such that $\mathbf{E}(\psi) \leq 2$ for all ψ . (*Hint:* proof of Tsirelson's bound involves rewriting the left-hand side in an enlightening way).
- (ii) *Research:* Suppose detector angles are configured as in (0.5). What can we infer about $|\psi\rangle$ from the value $\mathbf{E}(\psi)$? For instance, is there a meaningful relationship between $\mathbf{E}(\psi)$ and the entanglement entropy of $|\psi\rangle$? Indeed, we know the bound is saturated for at least one choice of maximally entangled $|\psi\rangle$; is it saturated for other Bell states, and if not, what are the detector angles required for saturation? Is there a property of $|\psi\rangle$ that determines whether $\mathbf{E}(\psi) > 2$ (e.g. entanglement)?⁴
- (iii) *Research:* Read up on Tsirelson's problem and its relation to Connes' embedding problem.

0.1. Aside: Recalibration of measurement apparatus.

The computational basis is the measurement basis determined by our apparatus. Our (logical) Pauli operators are defined with respect to this measurement basis.⁵ The apparatus measures any two-level quantum system, and corresponds to the measurement of some observable (a Hermitian operator on a two-dimensional state space); however, these details are forgotten when we move to the quantum circuit model: we say that we (*effectively*) measure spin along the z -axis (even though there may be no spin whatsoever). We might be able to 'recalibrate' our measurement apparatus, i.e. change the computational basis (Figure 1).

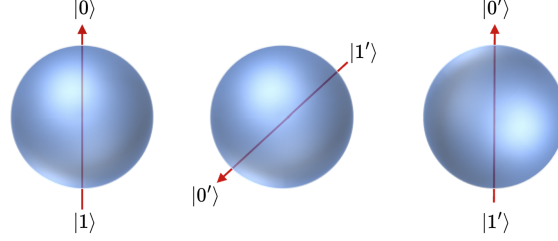


FIGURE 1. *Left:* Effective measurement of spin along z -axis. *Middle:* Recalibrate apparatus to measure spin along unit vector $\mathbf{v}$ (red arrow). *Right:* Treat $\mathbf{v}$ as the effective z -axis.

We will allow for more general n -qubit recalibrations in the following (i.e. can have an arbitrary change of computational basis on logical space; not just one that factors over n qubits). Now suppose that before the recalibration, we had some circuit $\mathcal{C}$. How do we realise the same circuit on our recalibrated system? There are two cases to consider: (A) the device that prepares the initial state does not change; (B) the device that prepares the initial state does so with the same amplitudes as before, but with respect to the new computational basis. I.e. find $\mathcal{C}'$ such that

$$\begin{aligned}
 \text{(A)} \quad & |\psi\rangle \xrightarrow{n} \boxed{\mathcal{C}} \text{---} \boxed{\text{Measurement}} = |\psi\rangle \xrightarrow{n} \boxed{\mathcal{C}'} \text{---} \boxed{\text{Measurement}}^{\mathcal{O}} \\
 \text{(B)} \quad & |\psi\rangle \xrightarrow{n} \boxed{\mathcal{C}} \text{---} \boxed{\text{Measurement}} = |\psi'\rangle \xrightarrow{n} \boxed{\mathcal{C}'} \text{---} \boxed{\text{Measurement}}^{\mathcal{O}}
 \end{aligned}$$

where $|\psi'\rangle = P|\psi\rangle$. As above, P is the matrix of whose columns are an orthonormal eigenbasis for $\mathcal{O}$, i.e. P is the change of basis transformation. Note that above, we used standard convention

$$\text{---} \boxed{\text{Measurement}} \text{---} = \text{---} \boxed{\text{Measurement}}^{Z^{\otimes n}} \text{---}$$

Case (A) is just the general n -qubit version of the case we studied above, i.e. just set $\mathcal{C}' = P^\dagger \mathcal{C}$.

Case (B). $\mathcal{C}'$ should evolve the amplitudes with respect to the new computational basis, in the exact same way that $\mathcal{C}$ did in the old basis. By standard linear algebra, it is clear that we must have $\mathcal{C}' = P\mathcal{C}P^\dagger$ in the above circuit, i.e.

⁴Chapter 4 ('Quantum Entanglement') of Preskill's notes for PH219 (2001) are illuminating. See e.g. Ch. 4.3.4.

⁵Suppose we have a spin-based qubit. If the apparatus measures spin along a unit vector $\mathbf{u}$, then the direction of $\mathbf{u}$ defines the positive z -axis. We conventionally set $|\uparrow_z\rangle = |0\rangle$ and $|\downarrow_z\rangle = |1\rangle$.

$$|\psi'\rangle \text{ --- } \boxed{\mathcal{C}'} \text{ --- } = |\psi'\rangle \text{ --- } \boxed{P^\dagger} \text{ --- } \boxed{\mathcal{C}} \text{ --- } \boxed{P} \text{ --- }$$

This has an intuitive interpretation. The $P^\dagger$ -gate sends $|\psi'\rangle$ to $|\psi\rangle$, we run circuit $\mathcal{C}$, and then the P -gate transforms back to the eigenbasis of $\mathcal{O}$. So on the left-hand side, if just before measurement we have state $\sum_i \lambda_i |a_i\rangle$ (where $\{|a_i\rangle\}_i$ is the computational basis), then on the right-hand side, just before measurement, we have state $\sum_i \lambda_i |b_i\rangle$ (where $\{|b_i\rangle\}_i$ is the eigenbasis of $\mathcal{O}$).

1-qubit case. Suppose we can perform a recalibration in the 1-qubit case, where the effective spin measurement is changed from being along $(0, 0, 1)^T$ to along $\mathbf{v}$ (as in Figure 1). In this instance, P is the matrix of orthonormal eigenvectors for $\mathbf{v} \cdot \boldsymbol{\sigma}$. $|0'\rangle$ corresponds to the $+1$ eigenvector, and $|1'\rangle$ corresponds to the -1 eigenvector.

Noisy measurement. We can flip case (A) on its head, and ask what happens if we have an imperfect measurement of a qubit: that is, the apparatus is perturbed and measures an observable $\mathcal{O}$ instead of Z , meaning a measurement is performed in a basis given by the columns of $P = P(\mathcal{O})$ (as opposed to the columns of I). We say the apparatus undergoes a perturbation P . In order for no error to result under a perturbation P , we know from the above that immediately before measurement, we must apply $|\psi\rangle \mapsto P^\dagger |\psi\rangle$. Thus, $\|P^\dagger - I\|$ quantifies the error (speaking – very loosely – for some suitable operator norm!). The noiseless average measurement is $\langle \psi | Z | \psi \rangle$, and the noisy average measurement is $\langle \psi | P^\dagger Z P | \psi \rangle$. Thus, the noise in our measured average can be quantified as $|\langle \psi | (P^\dagger Z P - Z) | \psi \rangle|$.

...and anyons. It is thus preferable for measurement bases not to lie in a continuum (one would envision that perturbing a basis is harder in this case). This is one of the many ways in which a topological quantum computer would be robust, where there are a finite number of measurement bases for any computation. If a computation involves k particles, a measurement basis is determined by the order in which particles are fused (the number of bases is equal to the $(k-1)^{th}$ Catalan number).

As an example, consider an Ising qubit. There are two measurement bases (a.k.a. fusion bases). As we discussed above, the default computational basis always corresponds to measurement of $\mathbf{v} \cdot \boldsymbol{\sigma}$, where $\mathbf{v} = (0, 0, 1)^T$. Any change of measurement basis for an Ising qubit has its change of basis matrix given by Hadamard matrix H . In terms of $\mathbf{v}$, this is $(0, 0, 1)^T \mapsto (1, 0, 0)^T$ (since the Hadamard matrix diagonalises the Pauli-X matrix). The upshot is that measurement is ‘rigid’ here, i.e. can only measure along z and x -axes, and moreover, the measurement apparatus cannot easily be perturbed between the two (since this corresponds to changing the order of fusion).

Even if our measurement bases lie in a discrete set – which precludes measuring ‘spin’ along some axis – we can usually (i.e. depending on the gates at our disposal!) mock-up such a measurement by transforming the quantum state immediately before measurement, as discussed above.

REFERENCES

- [1] Nielsen, M., and Chuang, I. (2010). "Quantum Computation and Quantum Information: 10th Anniversary Edition". *Cambridge University Press*.